# LECTURE 5

# Sound Waves, Velocity of sound in Solid, Liquid and Gases.

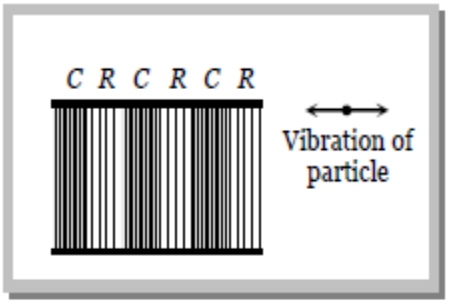


# Sound

**The energy to which human ears are sensitive**

## Sound Waves

- Mechanical Waves
- Travel through any material ( Solid, Liquid and gas)
- It Doesn't propagate in Space
- It is Longitudinal wave
- It travels in the form of compression and rarefaction



# What are the types of sound waves?

➢ **The categories cover different frequency ranges.**

**Audible waves**

- lie within the range of sensitivity of the human ear (20-20000) Hz. They can be generated in a variety of ways, such as by musical instruments, human voices, or loudspeakers.

**Infrasonic waves**

- have frequencies below the audible range (less than 20 Hz). Elephants can use infrasonic waves to communicate with one another, even when separated by many kilometers.

**Ultrasonic waves**

- have frequencies above the audible range (more than 20000 Hz). You may have used a “silent” whistle to retrieve your dog. Dogs easily hear the ultrasonic sound this whistle emits, although humans cannot detect it at all. Ultrasonic waves are also used in medical imaging.

# Speed of Sound Waves

**As Sound waves travel through air**

- elements of air are disturbed from their equilibrium positions.

Accompanying these movements are changes in

- Density.
- Pressure.

If the source of the sound waves vibrates sinusoidally, the density and pressure variations are also sinusoidal.

# The speed of sound varies from substance to substance

Sound travels most slowly in gases

It travels faster in liquids

and faster still in solids

- The speed of sound waves in a medium depends on the compressibility and the density of the medium.
- The compressibility can sometimes be expressed in terms of the elastic modulus of the material.
- The speed of all mechanical waves follows a general form:

$$v = \sqrt{\frac{elastic\ property}{internal\ property}}$$

**For solids**

$$v = \sqrt{\frac{Y}{\rho}}$$

**For fluids (Liquids and gases)**

$$v = \sqrt{\frac{B}{\rho}}$$

- $v$ is the speed of sound in a medium $\left(\frac{m}{s}\right)$.
- $Y$ is the Young's Modulus $\left(\frac{N}{m^2}\right)$.
- $B$ is the Bulk Modulus $\left(\frac{N}{m^2}\right)$.
- $\rho$ is the density of the medium $\left(\frac{Kg}{m^3}\right)$.

# Problem: 1

The Young's modulus for aluminum is $7.02 * 10^{10} \frac{N}{m^2}$. If the speed of sound in aluminum is measured to be $5.10\ km/s$, find its density (in $kg/m^3$).

## Sol.

$$Y = 7.02 * 10^{10} \frac{N}{m^2}\ , v = 5.10 \frac{km}{s} = 5.10 * 1000 = 5100 \frac{m}{s}$$

$$v = \sqrt{\frac{Y}{\rho}}$$

$$v^2 = \frac{Y}{\rho}$$

$$\rho = \frac{Y}{v^2} = \frac{7.02 * 10^{10}}{(5100)^2} = 2698.96 \frac{kg}{m^3} = 2.7 * 10^3 kg/m^3$$



# Problem: 2

The velocity of sound in sea water is $1.53 * 10^3 \frac{m}{s}$. Find the bulk modulus (in $\frac{N}{m^2}$) of sea water if its density is $1.03 * 10^3 \frac{kg}{m^3}$.

**Sol.**

$$v = \sqrt{\frac{B}{\rho}}$$

$$v^2 = \frac{B}{\rho}$$

$$B = v^2 . \rho$$

$$B = (1.53 * 10^3)^2 * 1.03 * 10^3$$

$$B = 2411127000 \frac{N}{m^2} = 2.41 * 10^9 \frac{N}{m^2}$$

# The Speed of Waves on Strings

$$\boldsymbol{v} = \sqrt{\frac{\boldsymbol{T}}{\mu}}$$

$T$ is the Tension $(N)$,

$where\ T = mg,$

$g$ is the gravitational acceleration $g = 9.8\ m/s^2$

$\mu$ is the mass per unit length $\left(\frac{Kg}{m}\right)$

# Problem: 2

A sculptor strikes a piece of marble with a hammer. Find the speed of sound through the marble (in $\frac{km}{s}$). (The Young's modulus is $50 * 10^9 \frac{N}{m^2}$ and its density is $2.7 * 10^3 \frac{kg}{m^3}$.)

**Sol.**

$$v = \sqrt{\frac{Y}{\rho}}$$

$$v = \sqrt{\frac{50 * 10^9}{2.7 * 10^3}}$$

$$v = 4303.3 \frac{m}{s}$$

$$v = 4.3 \frac{km}{s}$$

# Speed of Sound in Gases, Example Values

**TABLE 17.1** *Speed of Sound in Various Media*

| Medium | $v$ (m/s) | Medium | $v$ (m/s) | Medium | $v$ (m/s) |
|---|---|---|---|---|---|
| **Gases** | | **Liquids at 25°C** | | **Solids**[a] | |
| Hydrogen (0°C) | 1 286 | Glycerol | 1 904 | Pyrex glass | 5 640 |
| Helium (0°C) | 972 | Seawater | 1 533 | Iron | 5 950 |
| Air (20°C) | 343 | Water | 1 493 | Aluminum | 6 420 |
| Air (0°C) | 331 | Mercury | 1 450 | Brass | 4 700 |
| Oxygen (0°C) | 317 | Kerosene | 1 324 | Copper | 5 010 |
| | | Methyl alcohol | 1 143 | Gold | 3 240 |
| | | Carbon tetrachloride | 926 | Lucite | 2 680 |
| | | | | Lead | 1 960 |
| | | | | Rubber | 1 600 |

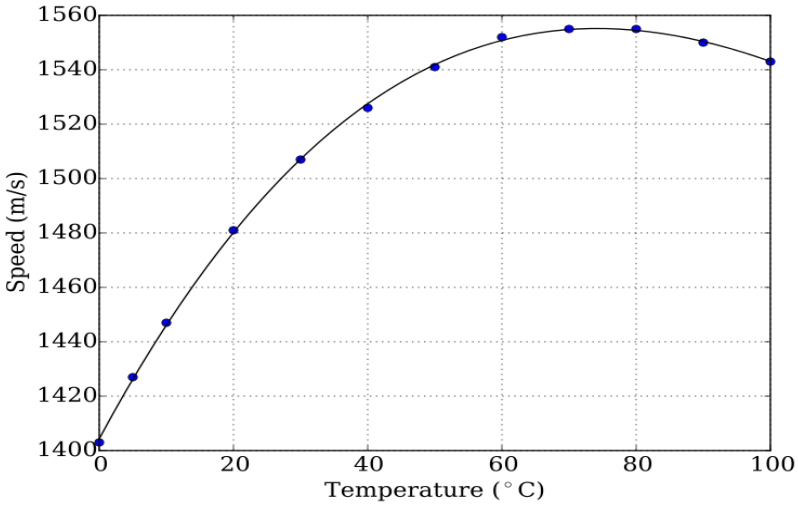
[a]Values given are for propagation of longitudinal waves in bulk media. Speeds for longitudinal waves in thin rods are smaller, and speeds of transverse waves in bulk are smaller yet.

# Speed of sound and the temperature

The speed of sound also depends on the temperature of the medium.
This is particularly important with gases.
For air, the relationship between the speed and temperature is

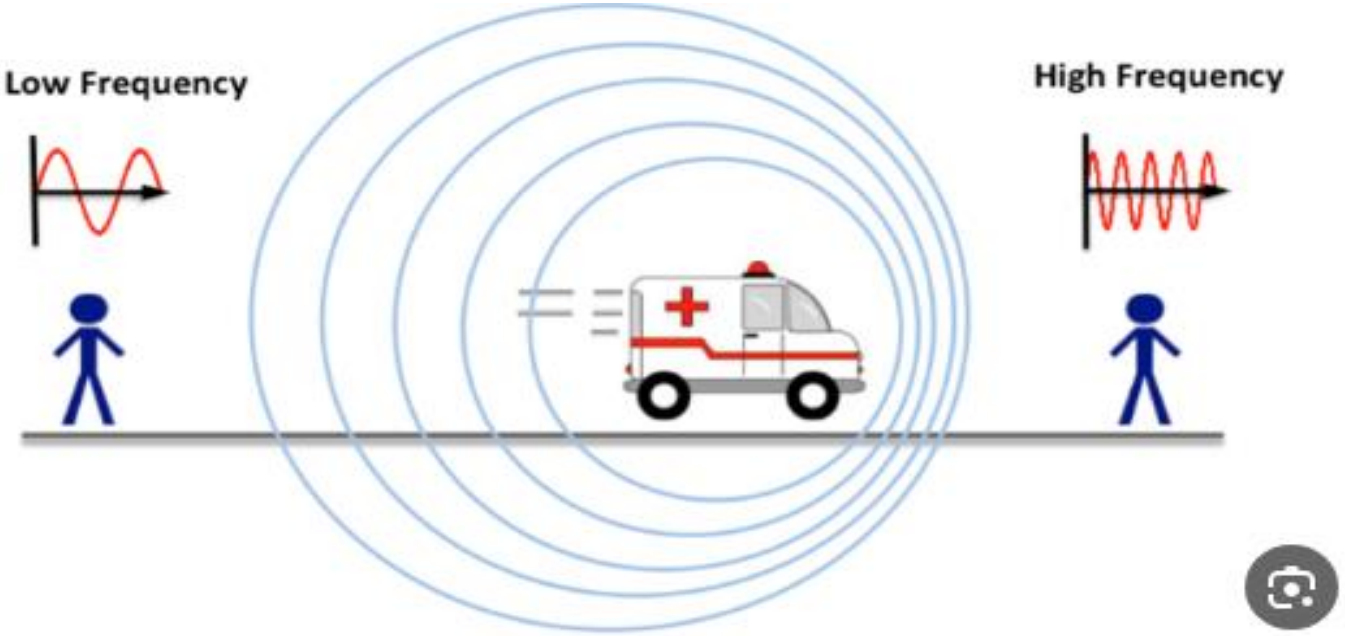
$$v = \left(331\frac{m}{s}\right)\sqrt{1+\frac{T_C}{273}}$$

The 331 m/s is the speed at $0^o C$.
$T_C$ is the air temperature in Celsius.



# Doppler effect:

Doppler effect is defined as the increase (or decrease) in the frequency of sound, light, or other waves as the source and observer move towards (or away from) each other.



# General Doppler Effect Formula (Sound Waves)

$$f' = \frac{(V \pm V_0)}{(V \pm V_s)} f$$

$f'$ = observed frequency

$f$ = actual frequency

$V$ = velocity of sound waves

$V_0$ = velocity of observer

$V_s$ = velocity of the source

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**Note on Signs:**

- Use **positive** $v_o$ when the observer is moving **toward** the source.
- Use **negative** $v_o$ when the observer is moving **away** from the source.
- Use **positive** $v_s$ when the source is moving **away** from the observer.
- Use **negative** $v_s$ when the source is moving **toward** the observer.

# Cases of Doppler Effect

## Case 1: Stationary Observer, Moving Source

When the observer is stationary ($v_o = 0$), the formula simplifies to:

$$f' = f\left(\frac{v}{v \pm v_s}\right)$$

- Use $v - v_s$ if the source is moving **toward** the observer (frequency increases).
- Use $v + v_s$ if the source is moving **away** from the observer (frequency decreases).

## Case 2: Moving Observer, Stationary Source

When the source is stationary ($v_s = 0$), the formula becomes:

$$f' = f\left(\frac{v \pm v_o}{v}\right)$$

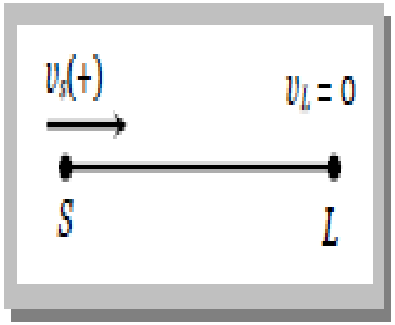
- Use $v + v_o$ if the observer is moving **toward** the source (frequency increases).
- Use $v - v_o$ if the observer is moving **away** from the source (frequency decreases).

# Sample problems based on Doppler effect

**Problem 42.** A source of sound of frequency 90 *vibration/sec* is approaching a stationary observer with a speed equal to 1/10 the speed of sound. What will be the frequency heard by the observer [MP PMT 2000]

(a) 80 *vibration/sec* (b) 90 *vibration/sec* (c) 100 *vibration/sec* (d) 120 *vibration/sec*

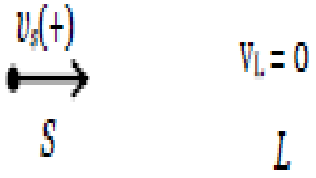
*Solution* : (c) $n' = \frac{v}{v - v_s} n \Rightarrow n' = \frac{v}{v - \frac{v}{10}} n \Rightarrow n' = \frac{10}{9} n = \frac{10 \times 90}{9} = 100$ vibration/sec




**Problem 43.** A source of sound of frequency 500 *Hz* is moving towards an observer with velocity 30 *m/s*. The speed of the sound is 330 *m/s*. The frequency heard by the observer will be [MP PET 2000]

(a) 550 *Hz* (b) 458.3 *Hz* (c) 530 *Hz* (d) 545.5 *Hz*

*Solution* : (a) $n' = \frac{v}{v - v_s} n \Rightarrow n' = \frac{330}{330 - 30} .500 \Rightarrow n' = 550$ *Hz*



# Applications of underwater sound

- Sonar
- Acoustic communication (Underwater communication)
- Acoustical oceanography


Thank
You